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Research Topic: Detecting and Mitigating Model Drift in Deployed Healthcare Machine Learning Systems: A Case Study of Sepsis Prediction Drift in ICU Data

Focus: Drafting Literature Review Outline, Methodology Layout, and Writing Chapter 4: Anticipated Outcomes

CHAPTER 2: LITERATURE REVIEW LAYOUT

2.1 Introduction to the Literature Review

This chapter provides a structured synthesis of existing literature on dataset shift and model drift in healthcare machine learning systems, with a specific focus on ICU-based sepsis prediction. Machine learning models are typically developed under the assumption that training and deployment environments are similar; however, healthcare systems are inherently dynamic. Changes in patient demographics, clinical protocols, and data recording practices introduce instability into predictive systems over time (*Finlayson et al., 2021; Guo et al., 2021*).

For example, a sepsis prediction model trained using ICU data from 2018 may rely heavily on laboratory testing patterns that change by 2022 due to updated clinical guidelines or resource constraints (*Ötleş et al., 2021*). As a result, the same model may produce systematically biased predictions when deployed in a new context.

This chapter therefore situates the research problem within broader theoretical and empirical debates, examining how such environmental changes undermine model stability (*Sahiner et al., 2023*). It further reviews global and context-relevant studies aligned with the study objectives and concludes by identifying gaps that motivate the proposed label-efficient drift monitoring framework.

2.2 Theoretical Background and Key Concepts

2.2.1 Dataset Shift and Concept Drift

Dataset shift refers to a change in the statistical properties of data between the training phase and deployment phase (*Quiñonero-Candela et al., 2008*). This can manifest in several forms:

- **Covariate shift:** Changes in input variables

  Example: A hospital introduces a new diagnostic device that records higher average lactate levels, shifting the distribution of a key sepsis predictor.

- **Prior probability shift:** Changes in outcome prevalence

  Example: During a pandemic, ICU admission rates increase, altering the proportion of sepsis cases (*Silva et al., 2025*).

- **Concept drift:** Changes in the relationship between inputs and outcomes

  Example: A new treatment protocol reduces mortality from sepsis, weakening previously strong predictors (*Gama et al., 2014*).

These shifts affect predictive performance because machine learning models assume that the joint distribution of inputs and outputs remains stable. When this assumption is violated, predictions become unreliable (*Gama et al., 2014; Quiñonero-Candela et al., 2008*).

2.2.2 Machine Learning in Healthcare Systems

Healthcare data presents unique challenges compared to other domains (*Finlayson et al., 2021*:

- **Heterogeneity:** Data comes from multiple sources (lab systems, electronic health records, monitoring devices)
- **Missingness:** Not all tests are conducted for all patients
- **Delayed labels:** Outcomes such as mortality or confirmed diagnosis may only be available after several days

Example: A sepsis model may require confirmation based on lab cultures, which may take 48–72 hours. During this period, real-time performance monitoring becomes difficult (*Ötleş et al., 2021*).

Additionally, healthcare systems are high-stakes environments. A false negative in sepsis prediction could delay treatment, while a false positive could lead to unnecessary interventions such as antibiotic overuse (*Subasri et al., 2025*).

2.2.3 Model Lifecycle Management

Traditional models are often deployed as static systems, meaning they are trained once and used indefinitely. However, in dynamic environments like healthcare, this approach is insufficient (*Guo et al., 2021*).

A dynamic lifecycle approach includes (*Silva et al., 2025*:

- Continuous monitoring of model inputs and outputs
- Periodic recalibration (adjusting predicted probabilities)
- Retraining with new data

Example: A hospital may deploy a sepsis model and monitor its performance monthly. If calibration deteriorates (e.g., predicted risk no longer matches observed outcomes), recalibration is performed before considering full retraining (*Guo et al., 2021*).

2.3 Review of Related Studies (Aligned to Objectives)

Objective 1: Detecting Model Drift in ICU Sepsis Prediction Systems

Studies have proposed several drift detection techniques (*Kore et al., 2024; Sahiner et al., 2023*:

- Statistical tests (e.g., Kolmogorov–Smirnov test) to compare distributions
- Population Stability Index (PSI) to quantify changes in feature distributions
- Monitoring prediction score distributions

Example: If the distribution of predicted sepsis risk suddenly shifts upward across all patients, this may indicate model instability (*Ötleş et al., 2021*).

However, healthcare studies show that drift is often subtle and context-dependent. A model may appear stable overall but fail for specific subgroups (e.g., elderly patients) (*Subasri et al., 2025*).

Objective 2: Evaluating the Impact of Drift on Model Performance and Reliability

Research shows that model performance degrades over time due to drift (*Guo et al., 2021; Ötleş et al., 2021*:

- Reduced accuracy and AUC scores
- Poor calibration (predicted probabilities do not match actual outcomes)
- Increased disparities across patient groups

Example: A sepsis model trained in one hospital may perform poorly in another due to differences in documentation practices or patient demographics (*Zech et al., 2018*).

This has practical implications: clinicians may lose trust in decision-support systems if predictions become inconsistent (*Finlayson et al., 2021*).

Objective 3: Developing and Assessing Mitigation Strategies

Common mitigation approaches include (*Guo et al., 2021; Silva et al., 2025*:

- **Recalibration:** Adjusting prediction thresholds or probability outputs

  Example: If a model consistently overestimates sepsis risk, calibration curves can be used to correct predictions.

- **Retraining:** Updating the model using new data

  Example: Incorporating recent ICU data to reflect updated clinical practices.

- **Domain adaptation:** Adjusting models for new environments

  Example: Adapting a model trained in one hospital for use in another (*Zech et al., 2018*).

Recent research also explores label-efficient methods, such as monitoring prediction distributions when outcome labels are unavailable (*Kore et al., 2024; Subasri et al., 2025*).

2.4 Summary of Identified Gaps

The literature reveals several persistent gaps (*Guo et al., 2021; Sahiner et al., 2023; Silva et al., 2025*:

- **Dependence on labeled data:** Many methods require real-time outcome labels, which are often delayed in ICU settings (*Ötleş et al., 2021*).
- **Context sensitivity of detection methods:** A method that works in one hospital may fail in another due to differences in patient populations (*Zech et al., 2018*).
- **Limited real-world validation:** Most studies rely on retrospective datasets rather than real deployment environments (*Kore et al., 2024*).
- **Lack of safe updating frameworks:** There are no standardized guidelines on when and how to update models safely (*Subasri et al., 2025*).
- **Weak governance integration:** Technical solutions often ignore clinical workflows and decision-making processes (*Finlayson et al., 2021*).

These gaps motivate the need for a label-efficient, governance-driven framework tailored to ICU-based sepsis prediction.

CHAPTER 3: METHODOLOGY LAYOUT

3.1 Research Design

This study adopts a quantitative, experimental, and longitudinal research design, aimed at evaluating how machine learning models degrade over time under real-world deployment conditions and how such degradation can be systematically detected and mitigated (*Guo et al., 2021; Silva et al., 2025*).

The quantitative approach enables objective measurement of model performance (e.g., discrimination, calibration), while the experimental component allows controlled simulation of drift scenarios. The longitudinal aspect is critical because model drift is inherently a temporal phenomenon, emerging as clinical environments evolve (*Gama et al., 2014*).

To approximate real-world deployment, the study uses temporal data partitioning, where models are trained on historical ICU data and evaluated on future data (*Ötleş et al., 2021*). For instance:

- Training set: ICU patient records from 2018–2019
- Validation set: Data from 2020
- Test (deployment simulation): Data from 2021–2022

This setup mimics a realistic scenario where a sepsis prediction model developed in 2019 is deployed in subsequent years. For example, if a hospital introduces a new early-warning protocol in 2021, the distribution of clinical features (e.g., frequency of lactate testing) may change, allowing the study to observe how such shifts impact model performance (*Finlayson et al., 2021*).

Additionally, synthetic drift scenarios may be introduced (e.g., artificially modifying feature distributions) to stress-test detection and mitigation strategies under controlled conditions (*Kore et al., 2024*).

3.2 Methodology by Objectives

Objective 1: Drift Detection

Data Sources and Acquisition

The study utilizes large-scale ICU datasets such as the MIMIC (Medical Information Mart for Intensive Care) database or equivalent clinical datasets. These datasets provide high-resolution, time-stamped patient information, enabling detailed temporal analysis (*Ötleş et al., 2021*).

Key variables include:

- **Vital signs:** heart rate, respiratory rate, blood pressure, oxygen saturation
- **Laboratory results:** lactate levels, white blood cell count, creatinine

- **Demographics:** age, gender, comorbidities
- **Clinical interventions:** antibiotic administration, fluid resuscitation

Example: A patient's lactate levels may be recorded multiple times per day, while blood pressure may be recorded continuously. These differences in measurement frequency are important because changes in recording practices themselves can introduce drift (*Finlayson et al., 2021*).

Data Cleaning and Preprocessing

Healthcare data is often incomplete and irregular, requiring rigorous preprocessing (Guo et al., 2021):

- **Missing value handling:**
  - Mean/median imputation for stable variables
  - Forward-filling for time-series data
  - Model-based imputation for complex patterns
- **Normalization and scaling:**
  - Ensures comparability across variables (e.g., standardizing lab values)
- **Temporal alignment:**
  - Aligning patient data into consistent time intervals (e.g., hourly bins)

Example: If heart rate is recorded every 5 minutes for one patient and every 30 minutes for another, both can be aggregated into hourly averages to ensure consistency.

Feature Engineering

Feature engineering transforms raw clinical data into meaningful predictors (Silva et al., 2025):

- **Temporal features:**
  - Rate of change in vital signs (e.g., increasing heart rate over 6 hours)
  - Variability measures (e.g., standard deviation of blood pressure)
- **Clinical scores:**
  - SOFA (Sequential Organ Failure Assessment) score
  - Early Warning Scores (EWS)
- **Derived indicators:**
  - Binary indicators (e.g., “lactate > 2 mmol/L”)
  - Interaction terms (e.g., age × comorbidity index)

Example: A sudden increase in lactate combined with declining blood pressure may be more predictive of sepsis than either feature alone.

Drift Detection Techniques

The study employs multiple complementary techniques (Kore et al., 2024; Sahiner et al., 2023):

- Kolmogorov–Smirnov (KS) Test:
  - Compares distributions of features between training and deployment periods
  - *Example:* Detecting whether lactate distributions in 2021 differ significantly from 2019
- Population Stability Index (PSI):
  - Quantifies magnitude of distribution change
  - PSI < 0.1 → no drift
  - PSI 0.1–0.2 → moderate drift
  - PSI > 0.2 → significant drift
  - *Example:* A PSI of 0.25 for white blood cell count suggests substantial shift
- Prediction Distribution Monitoring:
  - Tracks changes in predicted risk scores over time
  - *Example:* If average predicted sepsis risk increases from 15% to 30% across all patients, this may indicate model instability
- Subgroup Drift Analysis:
  - Detects drift within specific patient groups (Subasri et al., 2025)
  - *Example:* Model performance may degrade only for elderly patients while remaining stable overall

Objective 2: Performance Evaluation

Analytical Methods

Multiple models are used to ensure robustness (Guo et al., 2021):

- **Logistic Regression:**
  - Baseline, interpretable model widely used in clinical settings
- **Random Forest:**
  - Captures non-linear relationships and interactions
- **Gradient Boosting (e.g., XGBoost):**
  - High predictive performance, commonly used in healthcare ML

Example: A gradient boosting model may capture complex interactions between lab values and vital signs that logistic regression cannot.

Evaluation Metrics

- **AUC-ROC (Area Under Curve):**
  - Measures discrimination ability (ability to distinguish sepsis vs non-sepsis cases)
- **Calibration Curves:**
  - Assess how well predicted probabilities match actual outcomes
  - *Example:* If patients predicted at 80% risk only have 50% actual risk, the model is poorly calibrated (Ötleş et al., 2021)
- **Precision-Recall Curve:**
  - Particularly important for imbalanced datasets (sepsis is relatively rare)
- **Brier Score:**
  - Measures overall prediction accuracy (combines calibration and discrimination)

Temporal Validation

Models are evaluated using forward-in-time validation (Ötleş et al., 2021):

- Train on earlier data
- Test on later data

Example: Train: 2018–2019; Test: 2021. This approach captures real-world degradation. A model with AUC = 0.85 during training may drop to 0.75 in future data due to drift (Guo et al., 2021).

Objective 3: Drift Mitigation

Mitigation Techniques

- **Recalibration (Platt Scaling / Isotonic Regression):**
  - Adjusts predicted probabilities without changing model structure (Guo et al., 2021)
  - *Example:* If a model consistently overestimates risk, recalibration corrects probability outputs
- **Incremental Retraining:**
  - Incorporates new data into model training
  - *Example:* Updating the model using 2021 ICU data to reflect new clinical practices
- **Domain Adaptation:**
  - Adjusts models for new environments (Zech et al., 2018)
  - *Example:* Adapting a model trained in an urban hospital for use in a rural ICU with different patient characteristics

Validation Framework

- **Shadow Testing:**
  - New models run in parallel without affecting clinical decisions (Subasri et al., 2025)
  - *Example:* Comparing predictions from old vs updated model before deployment
- **Subgroup Analysis:**
  - Evaluating performance across demographics
  - *Example:* Ensuring the model performs equally well for male and female patients
- **Stress Testing:**
  - Simulating extreme drift scenarios
  - *Example:* Increasing prevalence of severe cases to test model robustness

3.3 Ethical Considerations

- Data confidentiality: Secure handling of patient records
- Bias mitigation: Ensuring equitable performance across groups (Subasri et al., 2025)
- Transparency: Clear documentation of model limitations

Example: If a model underpredicts sepsis risk for older patients, corrective measures must be implemented to avoid inequitable care (Finlayson et al., 2021).

3.4 Limitations of the Methodology

- Retrospective bias: Historical data may not reflect current practice
- Limited generalizability: Single dataset may not represent all hospitals (Zech et al., 2018)
- Simulation constraints: Cannot fully replicate real clinical workflows

Example: A model validated on MIMIC data may not perform identically in Ugandan ICU settings due to differences in infrastructure and clinical practices (Sahiner et al., 2023).

CHAPTER 4: ANTICIPATED OUTCOMES

4.1 Expected Findings

The study is expected to demonstrate that (Finlayson et al., 2021; Guo et al., 2021; Sahiner et al., 2023):

- Dataset shift is systematic, measurable, and clinically significant
- Performance metrics alone (e.g., AUC) may mask underlying drift
- Multi-signal monitoring enables earlier detection

Example: A model may maintain AUC = 0.82 but exhibit poor calibration, meaning predicted risks are misleading despite good ranking performance (Ötleş et al., 2021).

4.2 Contribution to Knowledge

- Provides empirical evidence on drift detection in high-stakes environments (Kore et al., 2024)
- Advances understanding of label-efficient monitoring approaches (Subasri et al., 2025)
- Bridges gap between theory and real-world deployment (Guo et al., 2021)

4.3 Practical Implications

- Hospitals can deploy real-time monitoring dashboards (Subasri et al., 2025)
- Clinicians gain more reliable decision-support tools (Finlayson et al., 2021)
- Health systems reduce risks of undetected model failure (Silva et al., 2025)

Example: A hospital dashboard may alert clinicians when model predictions become unstable, prompting review before harm occurs (Subasri et al., 2025).

4.4 Challenges and Mitigation Strategies

Challenge	Example	Mitigation
Limited labels	Sepsis confirmation delayed by lab tests	Use proxy indicators (prediction shifts, feature drift)
Model instability	Updated model performs worse in some groups	Use shadow deployment before rollout
Data variability	Different ICU protocols across hospitals	Apply domain adaptation techniques

Computational cost	Frequent monitoring is expensive	Use periodic sampling strategies
Bias risk	Lower accuracy for elderly patients	Continuous subgroup performance auditing

These challenges have been recognized across recent literature (*Guo et al., 2021; Kore et al., 2024; Sahiner et al., 2023; Silva et al., 2025; Subasri et al., 2025*).

In summary: This study addresses the critical challenge of model drift in deployed healthcare machine learning systems, with a specific focus on ICU-based sepsis prediction. Drawing on both foundational and recent literature, it establishes that dataset shift is not an exceptional occurrence but an inherent characteristic of real-world clinical environments. Changes in patient populations, clinical protocols, and data-generation processes continuously alter the conditions under which predictive models operate, thereby undermining their reliability over time.

The proposed methodological approach responds to this challenge by combining temporal validation, multi-signal drift detection, and structured mitigation strategies within a unified analytical framework. By incorporating both statistical detection methods and operational safeguards such as shadow testing and subgroup analysis, the study moves beyond purely technical solutions toward a more holistic, lifecycle-oriented perspective on model deployment.

Importantly, the anticipated outcomes highlight that reliance on traditional performance metrics alone is insufficient for maintaining model reliability. Instead, continuous monitoring, label-efficient detection strategies, and governed updating mechanisms are necessary to ensure sustained performance in high-stakes environments such as ICUs.

From an academic standpoint, the study contributes to ongoing discussions on dataset shift by grounding theoretical concepts in a clinically relevant use case. Practically, it offers a framework that can inform the design of monitoring systems and governance protocols for healthcare AI deployment. Ultimately, the study underscores the need to reconceptualize machine learning models not as static tools, but as dynamic systems requiring continuous oversight, adaptation, and validation to remain safe, effective, and equitable in real-world settings.

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